

The Extinction Probability

by Lisa Randall

If one knows the probability of a specified number of offspring being born in each generation, can one determine the extinction probability? One method of answering this question is the generating function, which incorporates the given values into an arbitrary polynomial.

Examples:

- (1) In a normal population, one might be given:
 $p(0) = 1/3$, $p(1) = 1/6$, $p(2) = 1/2$. So the generating function is $f(s) = 1/3 + (1/6)s + (1/2)s^2$.
- (2) On a subway at 5:00 PM, the number of passengers might increase according to $p(i) = 1/\sqrt{2}^{i+1}$, $i = 0, 1, 2, \dots$
so $f(s) = 1/2 + (1/4)s + (1/8)s^2 + \dots = 1/(2-s)$ ($|s| < 2$).
- (3) The probability for a given population of bees might be:
 $p(0) = 1/3$, $p(1) = 2/3$ so $f(s) = 1/3 + (2/3)s$

Note: The generating function has no meaning yet. We will find it of use later, when it is applied in finding a way to measure the extinction probability. Since $p(i)$ represents a probability, we know that:

- (4) $f(1) = \sum p(i) = 1$.
- (5) $f(0)$ is $p(0)$
- (6) We also know that in general,
 $f(s) = \sum p(i)s^i$.

Let $f^{(n)}(s)$ = generating function for the n -th generation probabilities and let $p(i)^n$ = probability of i individuals in the n -th generation.

We know that $f(s) = p(0) + p(1)s + p(2)s^2 + \dots$
For the second generation, $f^{(2)}(s) = p(1)^2s + p(0)(2)p(1)s + \dots$
One way of calculating the value of this expression would be to find the value of each of its members, eg:

$$p(0)(2) = p(0) + p(1)p(0) + p(2)p(0)^2 + p(3)p(0)^3 + \dots$$

It can be seen that this method would become too messy. (For those who disagree, calculate $p(17)^{(17)}$.)

An alternative and more useful approach would be:

$$f^{(2)}(s) = p(0) + p(1)(p(0) + p(1)s + p(2)s^2 + \dots) + p(2)(f(s))^2 + p(3)(f(s))^3 + \dots$$

We see that $f^{(2)}(s) = f(f(s))$ or, in general,

$$f^{(n)}(s) = f(f^{(n-1)}(s)) = \underbrace{f(f(\dots f(s)))}_{n \text{ times}}$$

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We now wish to find the extinction probability. We know that $f(0) = p(0)$ = probability of extinction by the first generation and that $p(0)^n$ = probability of extinction by the n -th generation.

$$f^{(2)}(0) = f(f(0)) = f(p(0)) = p(0)^2,$$

$$f^{(3)}(0) = f(f(f(0))) = p(0)^3$$

It can be seen that $p(0)^n$ is the limit of $f^{(n)}(0)$ as $n \rightarrow \infty$.

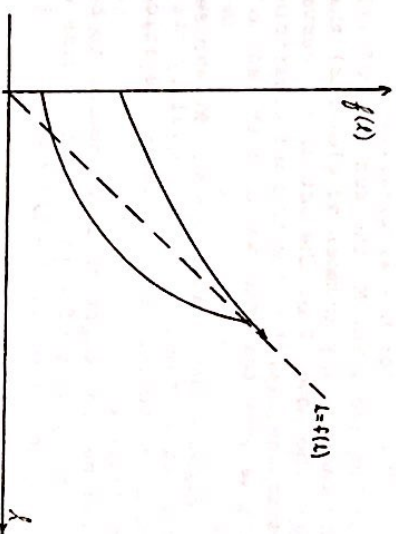
The completeness of real numbers implies that there exists an λ such that $\lim_{n \rightarrow \infty} p(0)^n = \lambda$ (the extinction probability), where the least upper bound of $p(0)^n$ is the limit. It has already been shown that $p(0)^n = f^{(n)}(0)$.

$$\lim_{n \rightarrow \infty} p(0)^n = \lim_{n \rightarrow \infty} f^{(n)}(0) = f(\lambda) = \lambda$$

so that $\lambda = f(\lambda)$.

This allows us to calculate the extinction probability for any set of given probabilities merely by substituting $\lambda = f(\lambda)$ into the equation. We refer to the previous examples:

- (1) $\lambda = 1/3 + (1/6)\lambda + (1/2)\lambda^2$ so $\lambda = 2/3$ or $\lambda = 1$. We use the lower value (least upper bound) so the extinction probability is $2/3$.
- (2) $\lambda = 1/(2-\lambda)$ so $\lambda = 1$ and the population will certainly go extinct.
- (3) is left to the reader.



One might note that λ is the value where $y = f(x)$ intersects the line $y = x$. In order for a population to survive, this function must intersect the line $y = x$ more than once. In other words, if the function approaches 1 from underneath, then it has already crossed the line once and the population has a chance of surviving. Conversely, if it approaches from above, extinction is certain. Therefore, by finding whether the slope of the function is greater or less than one as it approaches $(1,1)$, the fate of a given population can be ascertained.